

## EVERGLADES CITY 5 NE, FL, USA

WMO: 752040

Lat: 25.8996N

Lon: 81.3183W

Elev: 1

StdP: 101.31

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 92826

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>4.9 | (c)<br>7.2 | (d)<br>-0.6                   | (e)<br>3.6 | (f)<br>12.5 | (g)<br>2.0 | (h)<br>4.4 | (i)<br>11.7 | (j)<br>4.3            | (k)<br>18.3 | (l)<br>3.9 | (m)<br>18.6 | (n)<br>0.6            | (o)<br>N/A | (p)<br>0.274 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>8.6             | (c)<br>34.5     | (d)<br>25.7 | (e)<br>33.8 | (f)<br>25.5 | (g)<br>33.1 | (h)<br>25.4 | (i)<br>26.9         | (j)<br>31.8 | (k)<br>26.7 | (l)<br>31.5 | (m)<br>26.4 | (n)<br>31.0 | (o)<br>1.3           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>26.0                     | (b)<br>21.3 | (c)<br>28.1 | (d)<br>25.7 | (e)<br>20.9 | (f)<br>28.0 | (g)<br>25.3 | (h)<br>20.5 | (i)<br>28.0 | (j)<br>84.5   | (k)<br>31.9 | (l)<br>83.2 | (m)<br>31.3 | (n)<br>82.0 | (o)<br>31.1 | (p)<br>28.3       |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |      |      |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|------|------|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |      |      |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |      |      |
| (a) 3.8           | (b) 3.2 | (c) 2.9 | (e) 1.1                    | (f) 35.8 | (g) 2.5            | (h) 0.7 | (i) -0.7                                           | (j) 36.3 | (k) -2.2   | (l) 36.7 | (m) -3.6   | (n) 37.1 | (o) -5.5   | (p) 37.6 |      |      |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |      |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |      |
|                   |         |         |                            | 0.5      |                    | 27.7    | 2.4                                                | 0.4      | -1.1       | 28.0     | -2.5       | 28.2     | -3.8       | 28.4     | -5.6 | 28.7 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 24.1   | 17.9  | 20.0  | 20.7  | 23.5  | 25.5  | 27.6  | 28.4  | 28.7  | 28.2  | 26.0  | 22.0  | 20.7 |
|                                                                                  | DBStd       | 4.54   | 4.22  | 3.93  | 3.30  | 2.23  | 1.67  | 1.31  | 1.10  | 1.01  | 1.08  | 2.42  | 3.39  | 3.92 |
|                                                                                  | HDD10.0     | 4      | 3     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | HDD18.3     | 141    | 59    | 28    | 17    | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 9     | 25   |
|                                                                                  | CDD10.0     | 5167   | 249   | 281   | 333   | 406   | 481   | 529   | 572   | 581   | 545   | 497   | 362   | 332  |
|                                                                                  | CDD18.3     | 2263   | 47    | 75    | 92    | 158   | 223   | 279   | 313   | 323   | 295   | 239   | 121   | 97   |
|                                                                                  | CDH23.3     | 20361  | 309   | 546   | 796   | 1437  | 2048  | 2638  | 3126  | 3284  | 2763  | 2009  | 824   | 580  |
| CDH26.7                                                                          | 7841        | 44     | 127   | 206   | 538   | 897   | 1178  | 1352  | 1360  | 1064  | 725   | 221   | 129   |      |
| Wind                                                                             | WSAvg       | 1.1    | 1.1   | 1.2   | 1.3   | 1.3   | 1.2   | 1.0   | 0.9   | 0.9   | 1.0   | 1.2   | 1.2   | 1.1  |
| Precipitation                                                                    | PrecAvg     | 1341   | 41    | 41    | 47    | 45    | 92    | 267   | 200   | 205   | 223   | 101   | 44    | 36   |
|                                                                                  | PrecMax     | 1969   | 155   | 177   | 226   | 159   | 259   | 596   | 395   | 409   | 471   | 310   | 265   | 197  |
|                                                                                  | PrecMin     | 826    | 2     | 0     | 0     | 0     | 6     | 72    | 62    | 71    | 79    | 3     | 2     | 0    |
|                                                                                  | PrecStd     | 266    | 36    | 40    | 47    | 37    | 58    | 126   | 81    | 86    | 98    | 69    | 49    | 45   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 29.5  | 30.8  | 31.5  | 33.3  | 34.3  | 35.1  | 35.2  | 35.2  | 34.5  | 33.7  | 32.4  | 30.8 |
|                                                                                  |             | MCWB   | 22.3  | 22.7  | 22.5  | 23.5  | 23.5  | 25.7  | 26.3  | 26.4  | 26.0  | 25.4  | 24.6  | 23.8 |
|                                                                                  | 2%          | DB     | 28.0  | 29.5  | 30.1  | 32.0  | 33.1  | 34.1  | 34.1  | 34.2  | 33.5  | 32.5  | 30.6  | 29.3 |
|                                                                                  |             | MCWB   | 21.5  | 22.1  | 21.7  | 23.2  | 23.6  | 25.6  | 25.9  | 25.9  | 25.7  | 25.1  | 23.9  | 23.1 |
|                                                                                  | 5%          | DB     | 26.7  | 28.4  | 29.0  | 30.9  | 32.1  | 33.2  | 33.3  | 33.4  | 32.6  | 31.5  | 29.1  | 28.2 |
|                                                                                  |             | MCWB   | 20.6  | 21.6  | 21.3  | 22.6  | 23.6  | 25.4  | 25.8  | 26.0  | 25.6  | 24.6  | 22.9  | 22.4 |
|                                                                                  | 10%         | DB     | 25.2  | 27.1  | 27.8  | 29.9  | 31.2  | 32.2  | 32.5  | 32.6  | 31.8  | 30.5  | 27.8  | 26.8 |
|                                                                                  |             | MCWB   | 19.8  | 21.0  | 20.9  | 22.1  | 23.4  | 25.2  | 25.7  | 26.0  | 25.5  | 24.0  | 21.8  | 21.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.5  | 24.0  | 24.2  | 25.4  | 26.1  | 27.2  | 27.4  | 27.4  | 27.2  | 26.6  | 25.6  | 25.0 |
|                                                                                  |             | MCDB   | 27.6  | 28.5  | 28.7  | 31.1  | 31.9  | 32.6  | 33.0  | 32.3  | 31.5  | 30.0  | 29.7  | 28.9 |
|                                                                                  | 2%          | WB     | 22.6  | 23.2  | 23.5  | 24.8  | 25.3  | 26.7  | 26.8  | 26.9  | 26.8  | 26.1  | 24.9  | 24.1 |
|                                                                                  |             | MCDB   | 26.4  | 27.8  | 27.8  | 30.3  | 30.9  | 31.9  | 32.1  | 31.5  | 31.1  | 30.1  | 29.2  | 27.9 |
|                                                                                  | 5%          | WB     | 21.8  | 22.5  | 22.7  | 24.0  | 24.8  | 26.2  | 26.5  | 26.7  | 26.4  | 25.6  | 24.1  | 23.3 |
|                                                                                  |             | MCDB   | 25.3  | 27.0  | 27.1  | 29.1  | 30.0  | 31.2  | 31.4  | 31.2  | 30.7  | 29.5  | 28.1  | 27.0 |
|                                                                                  | 10%         | WB     | 20.8  | 21.7  | 21.8  | 23.3  | 24.2  | 25.8  | 26.2  | 26.4  | 26.1  | 25.1  | 23.3  | 22.4 |
|                                                                                  |             | MCDB   | 23.9  | 25.9  | 26.2  | 28.2  | 29.2  | 30.5  | 30.8  | 30.7  | 30.1  | 28.7  | 26.5  | 25.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.4  | 12.6  | 13.5  | 13.2  | 12.1  | 9.9   | 9.1   | 8.6   | 8.3   | 9.1   | 11.3  | 11.4 |
|                                                                                  |             | MCDBR  | 12.8  | 12.8  | 13.5  | 14.2  | 13.1  | 11.2  | 10.0  | 9.8   | 9.1   | 9.5   | 11.0  | 11.2 |
|                                                                                  |             | MCWBR  | 6.5   | 6.0   | 6.1   | 6.0   | 4.9   | 3.9   | 3.4   | 3.3   | 3.0   | 3.3   | 4.6   | 5.3  |
|                                                                                  | 5% WB       | MCDBR  | 11.0  | 11.7  | 11.7  | 12.3  | 11.4  | 10.1  | 9.6   | 9.0   | 8.4   | 8.7   | 10.0  | 10.1 |
|                                                                                  |             | MCWBR  | 6.4   | 5.9   | 6.1   | 5.6   | 4.6   | 3.8   | 3.5   | 3.3   | 3.1   | 3.2   | 4.3   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.355  | 0.370 | 0.382 | 0.396 | 0.425 | 0.452 | 0.462 | 0.454 | 0.426 | 0.406 | 0.370 | 0.359 |      |
|                                                                                  | taud        | 2.525  | 2.490 | 2.439 | 2.378 | 2.348 | 2.349 | 2.311 | 2.367 | 2.428 | 2.456 | 2.503 | 2.537 |      |
|                                                                                  | Ebn at Noon | 899    | 909   | 914   | 904   | 871   | 842   | 832   | 841   | 859   | 860   | 875   | 878   |      |
|                                                                                  | Edh at Noon | 93     | 103   | 114   | 124   | 127   | 126   | 131   | 123   | 113   | 105   | 94    | 89    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.77   | 4.63  | 5.62  | 6.41  | 6.48  | 5.84  | 5.83  | 5.55  | 5.02  | 4.71  | 4.06  | 3.52  |      |
|                                                                                  | RadStd      | 0.26   | 0.27  | 0.29  | 0.29  | 0.49  | 0.52  | 0.33  | 0.26  | 0.30  | 0.27  | 0.20  | 0.24  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (30 neighbors) |  | +0.39 | +0.91   | N/A    | N/A     | N/A   | N/A   | N/A         | -19     | +133    | +112    |

Nomenclature: See separate page